

Real-Time Structural Dynamics Digital Twin via IGA-POD

1. CAD & IGA Discretization

- Exact NURBS CAD Geometry
- C^{p-1} Continuity & High Fidelity
 - Full Order Model (FOM)
 $N = 12,450$ DOFs

Snapshots



2. Offline POD Reduction

- Snapshot Matrix SVD
- Optimal Mode Selection ($r = 12$)
 - Energy Ratio $\gamma > 99.999\%$
- Reduced Operators $\mathbf{M}_r, \mathbf{K}_r, \mathbf{C}_r$

$\mathbf{V}_r, \mathbf{M}_r$



3. Real-Time WebGL Twin

- Sub-millisecond step solver
- 386× Speedup over FOM
- WebGL 3D Interactive Render
 - Frame Rate > 24 FPS

Key Achievement: Sub-millisecond transient elastodynamics at 24+ FPS with relative error $< 10^{-4}$